

Algorithms and Data Structures

Problem Sheet 4

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36. Professor Gallup wants to create a random sample of the set $\{1, \dots, n\}$, that is is, an m -element subset S , where $1 \leq m \leq n$, such that each m -subset is equally likely to be created. One way is to set $A[i] = i$, for $i = 1, 2, \dots, n$, call **RANDOMLY-PERMUTE(A)**, and then take just the first m array elements. Show that this indeed produces a random sample.
37. The method from Exercise 36 makes n calls to the **RANDOM** procedure. In Professor Gallup's application, n is much larger than m , and so the professor wants to create a random sample with fewer calls to **RANDOM**.

```
1: procedure RANDOM_SAMPLE( $m, n$ )
2:    $S = \emptyset$ 
3:   for  $k = n - m + 1$  to  $n$  do
4:      $i = \text{RANDOM}(1, k)$ 
5:     if  $i \in S$  then
6:        $S = S \cup \{k\}$ 
7:     else
8:        $S = S \cup \{i\}$ 
9:     end if
10:  end for
11:  return  $S$ 
12: end procedure
```

Show that the procedure **RANDOM_SAMPLE** returns a random m -subset S of $\{1, \dots, n\}$.

38. Professor Knieval suggests the procedure **PERMUTE-BY-CYCLE** to generate a uniform random permutation. Show that each element $A[i]$ has a $1/n$ probability of winding up in any particular position in B . Then show that Professor Knieval is mistaken by showing that the resulting permutation is not uniformly random.

```
1: procedure PERMUTE-BY-CYCLE( $A, n$ )
2:   let  $B[1 : n]$  be a new array
3:    $offset = \text{RANDOM}(1, n)$ 
4:   for  $i = 1$  to  $n$  do
5:      $dest = i + offset$ 
6:     if  $dest > n$  then
```

```

7:         dest = dest - n
8:     end if
9:     B[dest] = A[i]
10: end for
11: return B
12: end procedure

```

39. Professor Kelp decides to write a procedure that produces at random any permutation except the identity permutation, in which every element ends up where it started. He proposes the procedure PERMUTE-WITHOUT-IDENTITY. Does this procedure do what Professor Kelp intends?

```

1: procedure PERMUTE-WITHOUT-IDENTITY(A,n)
2:   for i = 1 to n - 1 do
3:     swap A[i] with A[RANDOM(i + 1, n)]
4:   end for
5: end procedure

```

40. Consider the PERMUTE-WITH-ALL, which instead of swapping element $A[i]$ with a random element from the subarray $A[i : n]$, swaps it with a random element from anywhere in the array. Does PERMUTE-WITH-ALL produce a uniform random permutation? Why or why not?

```

1: procedure PERMUTE-WITH-ALL(A,n)
2:   for i = 1 to n do
3:     swap A[i] with A[RANDOM(1, n)]
4:   end for
5: end procedure

```

41. What are the minimum and maximum numbers of elements in a heap of height h ? Show that an n -element heap has height $\lfloor \lg n \rfloor$.
42. Consider a max-heap, in which all elements are distinct. Where can the smallest element reside? At which levels in a max-heap might the k th largest element reside, for $2 \leq k \leq \lfloor \frac{n}{2} \rfloor$?
43. Is an array that is in sorted order a min-heap? Is the array with values

[33, 19, 20, 15, 13, 10, 2, 13, 16, 12]

a max-heap?